



Retinal Review Workbench

Clinician User Manual

Clinician-led retinal review and dataset preparation

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Research and clinical-support proof of concept for public or synthetic data. Model output is assistive evidence; human review remains authoritative.

Table of contents

1 Quick Start	2
2 Before You Begin	3
2.1 Intended users and scope	3
2.2 Where the images come from	3
2.3 Supported images	3
2.4 Model-assisted evidence	3
2.5 Workstation areas	3
3 Worklist and Confirm Image	4
3.1 Open the Worklist	4
3.2 Confirm Image	5
3.3 Admission exceptions	5
4 Review and Optional AI Assistance	6
4.1 Inspect the image	6
4.2 Read the model suggestions	6
4.3 Show or filter the overlays	7
5 Confirm DR Grade	7
6 Edit Annotations	8
6.1 Add human annotations	8
6.2 Correct or remove an AI ROI	9
7 Working Through Cases	10
7.1 Reviewer default	10
7.2 Previous / Next and Save & Next	11
7.3 Autosave and unsaved changes	11
7.4 Next action hint	11
8 Confirm Annotation	12
9 Dataset Readiness and Export	12
9.1 Check readiness	12
9.2 Export the manifest	13
10 Models & Audit and Traceability	14
10.1 Models & Audit	14
10.2 What the workstation records	14
10.3 Optional CVAT	15
11 Troubleshooting, Safety, and Glossary	15
11.1 Troubleshooting	15
11.2 Safety and privacy	15
11.3 Glossary	16

1 Quick Start

Retinal Review Workbench takes one admitted retinal image through a controlled review and turns the clinician's decisions into a traceable dataset record. Use this page as the everyday checklist; the following chapters explain each step.

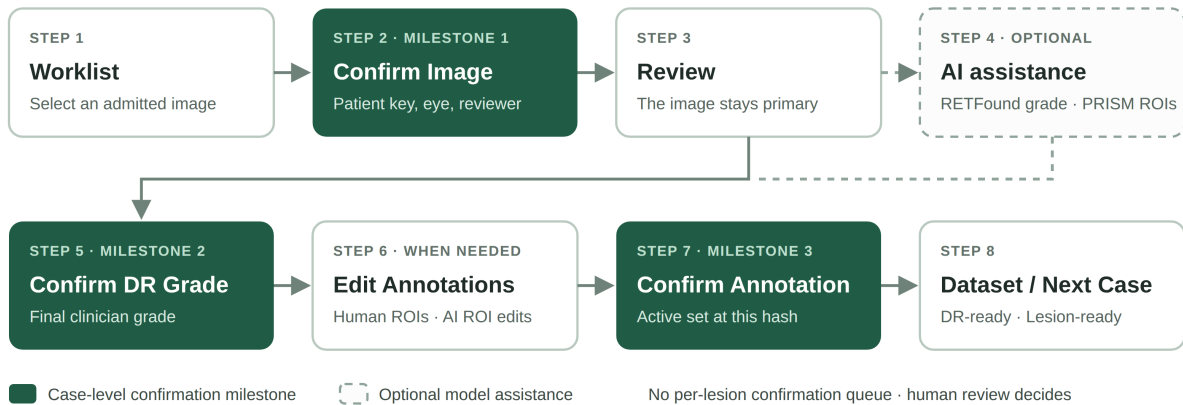


Figure 1.1: The clinician workflow. The three green boxes are the case-level confirmation milestones; AI assistance is optional.

1. **Worklist** – select the case. Open the row menu (...) and choose **Confirm Image**.
2. **Confirm Image** – check the pseudonymous patient key, the eye, and your reviewer name, then choose **Confirm Image**.
3. **Review** – inspect the retinal image. Choose **Analyze** only when optional model assistance is useful and available.
4. **Optional AI** – read the RETFound grade suggestion and toggle the PRISM lesion overlays with **Show AI suggestions**.
5. **Clinician Review** – select the **Final DR grade** and choose **Confirm DR Grade**.
6. **Edit annotations** – only when a human annotation, or a correction or removal of an AI ROI, is needed.
7. **Confirm Annotation** – confirm the active annotation set after reviewing it.
8. **Datasets** – check **DR-ready** and **Lesion-ready** separately, then export or move to the next case.

Important – three confirmations

Confirm Image, **Confirm DR Grade**, and **Confirm Annotation** are the only case-level confirmations. AI suggestions never need to be accepted or rejected one by one before the review can be completed.

2 Before You Begin

2.1 Intended users and scope

This manual is for clinicians and reviewers using the review workstation. Workstation installation, the hospital Model API, and backups are covered in the separate *Deployment and Operations Manual*.

Retinal Review Workbench is a clinician-controlled research and clinical-support workstation. It keeps source identity, patient and eye context, human decisions, annotations, model evidence, and export provenance together for review. It does not establish a diagnosis, disease truth, referral decision, or treatment decision. The final clinician review remains authoritative.

2.2 Where the images come from

You, or another authorized hospital user, choose each image from hospital-controlled storage, and the approved hospital process stages it into the Workspace input folder. The project team does not browse hospital storage or pre-select images. It monitors overall Refer/Not-refer, sex, and left/right coverage, so you do not manage dataset quotas yourself. What to choose, what to avoid, and what to do when unsure are covered in the one-page *Clinician Image Selection Guide* (docs/CLINICIAN_IMAGE_SELECTION_GUIDE.pdf).

2.3 Supported images

The workstation accepts JPEG, PNG, TIFF, and supported single-frame ophthalmic fundus DICOM. MRI, OCT, PACS objects, non-fundus DICOM, unsupported SOP classes, multi-frame objects, and untested transfer syntaxes are outside model support. A recognized non-fundus DICOM is shown as **Unsupported modality**, not as a generic read failure.

2.4 Model-assisted evidence

- **RETFound** (retfound-aptos5) provides a five-class DR grade suggestion.
- **PRISM-DR** (prism-dr-5fold) provides optional lesion ROI suggestions.

A displayed confidence or score is a model score, not a calibrated clinical probability. An empty PRISM result does not prove that lesions are absent.

2.5 Workstation areas

Area	What you do there
Worklist	Find admitted cases, see their status, and start Confirm Image .
Review	Inspect the image and optional model assistance (read-only).
Clinician Review	Record the final DR grade, or escalate an exception.
Edit annotations	Draw human annotations; correct or remove an AI ROI when needed.
Datasets	Check DR-ready and Lesion-ready status and export the manifest.
Models & Audit	Read model, inference, hash, and annotation provenance.
Settings	Select a Workspace, set the default reviewer, and manage hints.

Safety – human authority

Treat every model output as assistive evidence. Do not use a model score or an empty lesion result as a diagnosis, a referral decision, or proof that disease is absent.

3 Worklist and Confirm Image

3.1 Open the Worklist

Purpose. Start a review session in the approved Workspace.

Steps.

1. Start the workstation with the launcher provided by your operator and open `http://127.0.0.1:8000/app/`.
2. Check that the sidebar shows the correct **Active workspace**. If not, open **Settings** and select it.
3. In **Worklist**, choose **Scan input folder** when new images have been added.

Expected result. Each admitted image appears as one row with its patient and eye context, AI result, review status, and actions. Source files are never moved or changed.

Worklist

Review admitted images, resolve patient and eye identity, and keep AI assistance separate from the clinician decision.

Filters

Sort: Filename

Cases

Patients

Search by filename or pseudonymous patient key.

7 cases - 5 need attention

IMAGE	PATIENT / EYE	AI RESULT	REVIEW	ACTION
 SYNTH_001.png	Patient not linked Eye side needs confirmation Link patient and confirm eye	Grade 2 Suggestion available	Pending review	Review ...
 SYNTH_002_R.png	PAT0002 · Right Confirmed 	Not analyzed	Reviewed	Review ...
 SYNTH_003_L.png	PAT0003 · Left Confirmed 	Not analyzed	Reviewed	Review ...
 SYNTH_004_R.png	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Review ...

Figure 3.1: Worklist with synthetic cases. Confirmed rows show the patient key and eye; unconfirmed rows offer Link patient and confirm eye.

3.2 Confirm Image

Purpose. Confirm the image context before analysis or clinical review. This is the first confirmation milestone.

Steps.

1. In the case row, open the menu (...) and choose **Confirm Image**.
2. Check the filename and the note above the fields.
3. Enter the **Pseudonymous patient key**, for example PAT0001. Leave it empty when the case should stay unlinked.
4. Set **Eye** to **Left**, **Right**, or **Unknown**.
5. Enter your **Reviewer name**. Select **Use as default reviewer on this workstation** to prefill it next time.
6. Choose **Confirm Image**.

Expected result. The patient and eye context, reviewer, and timestamp are saved, and the row shows **Confirmed**. The case can proceed to **Review**.

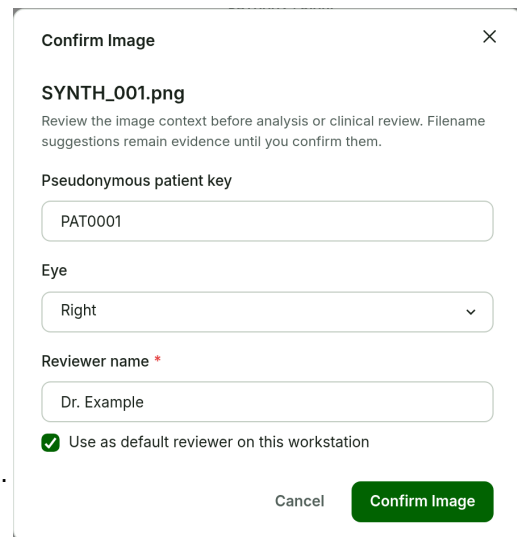
A screenshot of the 'Confirm Image' dialog box. The title bar says 'Confirm Image' with a close button (X). The main content area shows 'SYNTH_001.png' at the top. Below it is a note: 'Review the image context before analysis or clinical review. Filename suggestions remain evidence until you confirm them.' There are three input fields: 'Pseudonymous patient key' with the value 'PAT0001', 'Eye' with a dropdown menu showing 'Right', and 'Reviewer name' with the value 'Dr. Example'. Below the reviewer name field is a checkbox labeled 'Use as default reviewer on this workstation' which is checked. At the bottom right are two buttons: 'Cancel' and 'Confirm Image' (which is green).

Figure 3.2: Confirm Image dialog.

Filename or resolver suggestions are evidence only; nothing is inferred from a filename until you confirm it.

3.3 Admission exceptions

Use the row status to decide the next action. **Resolve image readiness** and **Exclude from queue** are available in the same row menu. A recognized MRI or other non-fundus DICOM stays **Unsupported modality** and is never sent to RETFound or PRISM-DR.

Important – source files

Do not rename, move, overwrite, or recompress admitted source images. The workstation records the original source identity and works from it.

4 Review and Optional AI Assistance

4.1 Inspect the image

Purpose. Inspect the retinal image and decide whether optional model evidence is useful.

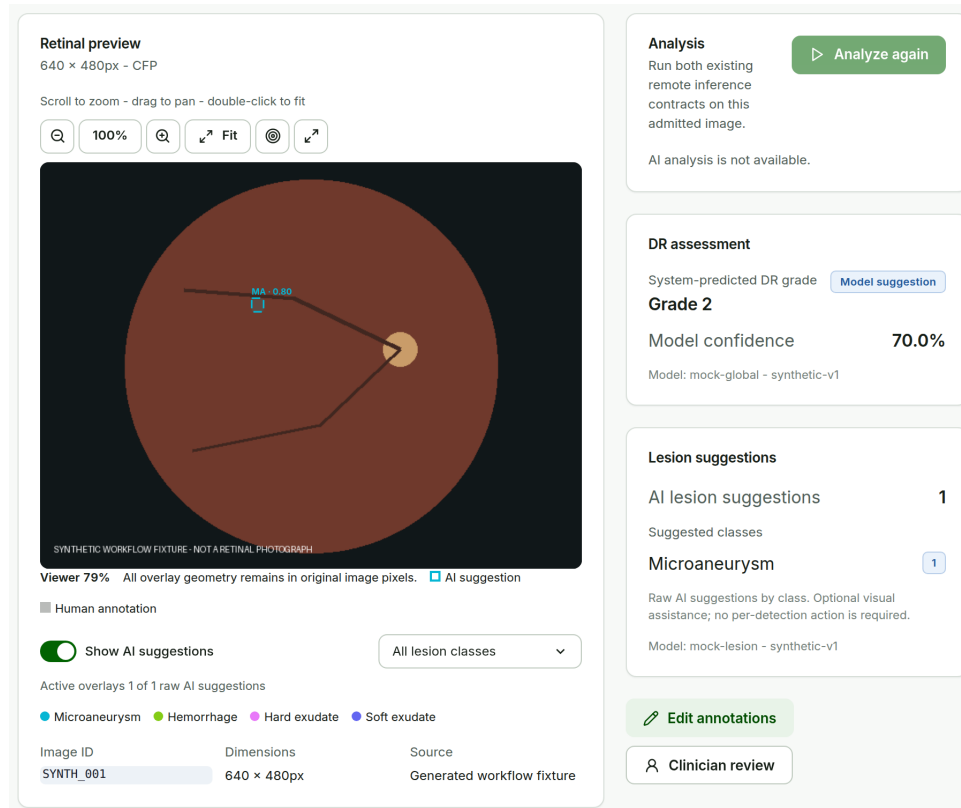


Figure 4.1: Review page for the synthetic case SYNTH_001: image on the left, model assistance on the right.

Steps.

1. Open the case with **Review** from the Worklist.
2. Use the viewer controls to zoom, fit, pan, inspect coordinates, or open the full-screen view.
3. Check the **Patient/Eye** line above the image.
4. Choose **Analyze** when model assistance is wanted and the Model API is available. If it is not available, continue with manual review.

Expected result. The retinal image stays the main surface. The right column shows **Analysis**, **DR assessment**, and **Lesion suggestions**, followed by **Edit annotations** and **Clinician review**.

4.2 Read the model suggestions

- **DR assessment** shows the RETFound grade suggestion and its model confidence. This is a model score, not a calibrated probability.
- **Lesion suggestions** counts PRISM-DR suggestions by class. An untouched overlay is labelled with class and score, for example MA · 0.80.
- The lesion summary is not a task list. The review can be completed without confirming, rejecting, or reducing any AI detection.
- Review is read-only. Corrections are made in **Edit annotations** (see Section 6).

4.3 Show or filter the overlays

Use **Show AI suggestions** to show or hide the PRISM overlays, and the lesion filter to display **All lesion classes** or one class: **MA · Microaneurysm**, **HE · Hemorrhage**, **EX · Hard exudate**, or **SE · Soft exudate**. The counter shows how many raw AI suggestions are active.



Figure 4.2: Overlay controls below the viewer.

5 Confirm DR Grade

Purpose. Record the clinician's final DR grade. This is the second confirmation milestone.

Steps.

1. From **Review**, choose **Clinician review**.
2. Check **Reviewer name**. It is prefilled when a workstation default is set.
3. Read the **AI suggestion** line, if a suggestion exists.
4. Select the **Final DR grade** (Grade 0–4).
5. Add a **Remark** when useful.
6. Choose **Confirm DR Grade**. Choose **Escalate** when the case needs a separate decision.

Expected result. A green confirmation appears and **Review record** shows the action, final grade, and remark. Keeping the model grade is recorded as AI_ACCEPTED, a different grade as AI_CORRECTED, and a grade without a model result as MANUAL. The human grade is the authoritative decision.

Clinician sign-off
Save a human review action with an optional remark.

Reviewer name *

Dr. Example

☒ Use as default reviewer on this workstation

AI suggestion: Grade 2

Final DR grade

Grade 2

Remark

Synthetic training case; grade agrees with suggestion.

Confirm DR Grade

Escalate

DR grade confirmed from the AI suggestion for SYNTH_001.

Confirm DR Grade

Escalate

Review record
Persisted in the existing SQLite case store.

Action: ACCEPT

Final grade: 2

Remark: Synthetic training case; grade agrees with suggestion.

Revision 4

Figure 5.2: The saved review record.

Figure 5.1: Clinician sign-off before confirming.

Confirm DR Grade & Next at the top of the page saves the same decision and opens the next case.

6 Edit Annotations

6.1 Add human annotations

Purpose. Record human annotations, or change an AI ROI, only when the clinician identifies a change that should be recorded.

Steps.

1. From **Review** or **Clinician Review**, choose **Edit annotations**. The page opens as **Annotation Editor**.
2. Under **Editor tools**, choose **Box**, **Polygon**, **Point**, or **Circle**, then choose the **Lesion class**.
3. Draw on the image. Double-click to finish a polygon.
4. Use **Select**, **Undo**, **Delete selected**, and **Lock selected** to adjust your own annotations.
5. Keep **AI suggestions** shown when comparing with the model, or hide them to reduce clutter.

Expected result. Human annotations are stored in original image pixels and remain separate from the AI evidence. The draft is saved automatically (see Section 7.3).

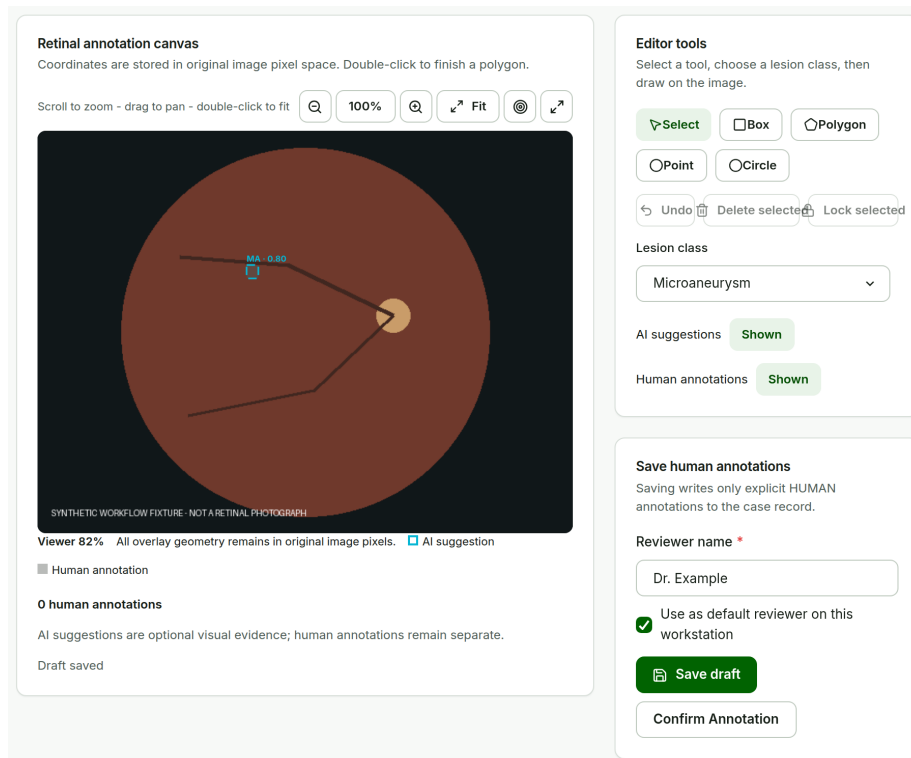


Figure 6.1: Annotation Editor: canvas, editor tools, and the save panel.

6.2 Correct or remove an AI ROI

Purpose. Fix an obvious model class, or remove a wrong detection. This is optional and is not a confirmation queue.

Steps.

1. In the Annotation Editor, keep **AI suggestions** shown and click the AI ROI on the image.
2. In **Optional ROI action**, read the original **AI suggestion** with its class and score.
3. To correct the class, choose a new value in **Change lesion class** and choose **Save class correction**.
4. To discard the detection, choose **Remove this detection**.
5. Check **Reviewer name** before saving.

Expected result. The corrected ROI shows the new class as **clinician corrected**, without a score. The original AI class, score, geometry, and provenance stay available for audit. A removed detection leaves the active overlay but remains in the audit record.

Important – score meaning

A model score belongs to the class the model predicted. If PRISM suggests Hemorrhage · 0.90 and you change the class to Microaneurysm, the result is **Microaneurysm – clinician corrected**. It is never shown as Microaneurysm · 0.90.

Dense geometry work can use the optional CVAT round trip (Section 10.3). The Annotation Editor does not replace it.

Optional ROI action [X]

AI suggestion: **Microaneurysm** · 0.80

Change lesion class

Microaneurysm [v]

Reviewer name *

Dr. Example

☒ Use as default reviewer on this workstation

Save class correction

Remove this detection Cancel

Selected: MA - 0.80

Figure 6.2: Selected AI ROI with its popover.

Optional ROI action [X]

AI suggestion: **Microaneurysm** · 0.80

Current reviewed class: Hemorrhage. The AI score remains attached to the original class only.

Change lesion class

Hemorrhage [v]

Reviewer name *

Dr. Example

☒ Use as default reviewer on this workstation

Save class correction

Remove this detection Cancel

Selected: HE - clinician corrected

Figure 6.3: After correction: the score stays with the original class.

7 Working Through Cases

These conveniences reduce repetitive work. None of them confirms a milestone or changes clinical state on its own.

7.1 Reviewer default

Select **Use as default reviewer on this workstation** in any reviewer field, or set **Default reviewer** in **Settings**, to prefill new or unattributed review forms.

Review preferences

Set a workstation default for new review records. It is editable and is not an authenticated identity.

Default reviewer

Dr. Example

New or unattributed review forms use this value. Clear it for a blank form; previously saved reviewers are unchanged.

☒ Show next action hints

Hints are guidance only. They do not change clinical state and are stored on this workstation.

Figure 7.1: Review preferences in Settings.

The value is stored only in this browser. It stays editable, it is not authentication, and it never rewrites reviewer names already recorded. On a shared workstation, change or clear it when the user changes.

7.2 Previous / Next and Save & Next

Previous and **Next** move through the case order shown in the middle (for example 1 of 7). Moving does not save or confirm anything. In the Annotation Editor, **Save & Next Case** saves the current draft and opens the next case; in Clinician Review, **Confirm DR Grade & Next** confirms the grade first.

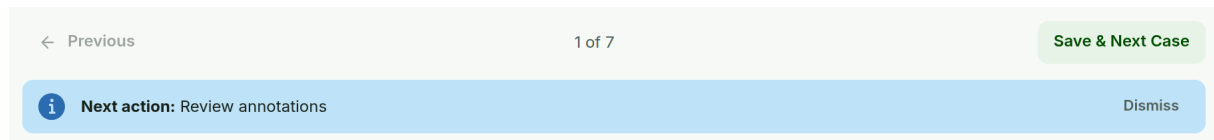


Figure 7.2: Case navigation and the Next action hint at the top of each case page.

7.3 Autosave and unsaved changes

In the Annotation Editor, edits are saved as a draft about one second after you stop editing, provided a reviewer name is entered. The status below the canvas shows **Unsaved changes**, **Saving draft...**, **Draft saved**, or **Save failed — retry**. **Save draft** saves immediately.

1 human annotation

AI suggestions are optional visual evidence; human annotations remain separate.

Draft saved

Figure 7.3: Draft status after one human annotation was added.

A draft is not a confirmation. If you try to leave while changes are unsaved, saving, or failed, the workstation asks *You have unsaved annotation changes. Leave this case?* Choose **Cancel** to stay and save.

7.4 Next action hint

The blue **Next action** line suggests the next step, for example **Confirm image**, **Confirm DR grade**, **Review annotations**, or **Complete**. It is guidance only. Choose **Dismiss** to hide hints on this workstation; turn them back on with **Show next action hints** in **Settings**.

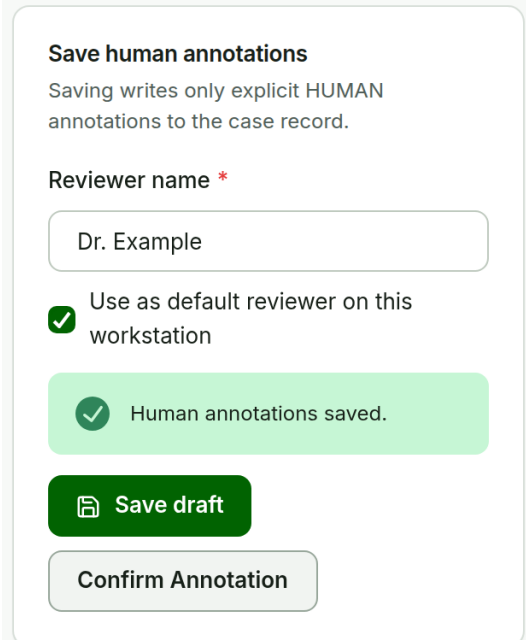
8 Confirm Annotation

Purpose. Confirm that the current active annotation set has been reviewed. This is the third confirmation milestone.

Steps.

1. Finish any human annotations or AI ROI changes.
2. Check **Reviewer name** in **Save human annotations**.
3. Choose **Confirm Annotation**.

Expected result. The draft is saved, then the reviewer, timestamp, and a hash of the active set are recorded. The **Next action** hint changes to **Complete** when the grade is also confirmed.



Save human annotations
Saving writes only explicit HUMAN annotations to the case record.

Reviewer name *

Dr. Example

☒ Use as default reviewer on this workstation

☒ Human annotations saved.

 Save draft

Confirm Annotation

Figure 8.1: Save human annotations after Confirm Annotation.

Later edits. Any later edit changes the hash, so the case needs **Confirm Annotation** again before it is Lesion-ready. Confirming an empty set is allowed; it records that the set was reviewed, not that lesions are absent.

9 Dataset Readiness and Export

9.1 Check readiness

Purpose. See which cases are ready for each task. Classification readiness and lesion readiness are separate states.

Steps.

1. Open **Datasets**. The summary line shows **Images**, **Annotations**, **DR-ready**, and **Lesion-ready** for the active Workspace.
2. Choose a tab: **All**, **DR-ready**, **Lesion-ready**, **Needs review**, or **Excluded**.
3. Read the two status chips in **Dataset status**: one for DR, one for lesions.
4. Fix a case in **Clinician Review** or **Edit annotations** when its status asks for it.

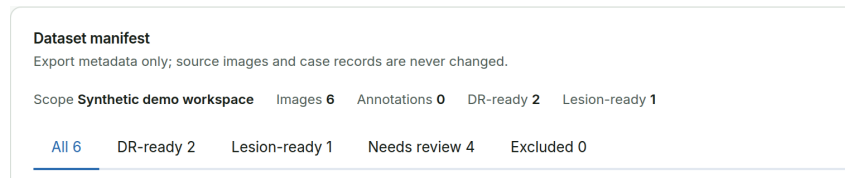


Figure 9.1: Dataset manifest summary with separate DR-ready and Lesion-ready counts.

PATIENT / EYE	CLINICIAN GRADE	ANNOTATIONS	DATASET STATUS
PAT0002 / Right	Grade 1	None	DR: Ready Lesions: Ready
PAT0003 / Left	Grade 0	None	DR: Ready Lesions: NO ANNOTATION CONFIRMATION

Figure 9.2: The DR-ready tab. PAT0002 is both DR-ready and Lesion-ready; PAT0003 is DR-ready but still needs annotation confirmation.

Expected result.

- **DR-ready** needs valid source provenance, an included fundus image, and a final clinician grade with reviewer and timestamp.
- **Lesion-ready** needs valid source provenance, an included fundus image, and a current **Confirm Annotation** whose hash matches the active set.

One task can be ready while the other is not. AI-only evidence is never promoted to ground truth by itself.

9.2 Export the manifest

Steps.

1. Check the readiness of the active Workspace.
2. Choose **Export manifest**.
3. Read the confirmation message.

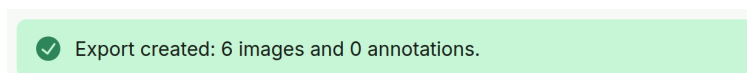


Figure 9.3: Export confirmation.

Expected result. The package is written to the Workspace output folder: `manifest.json`, `images.csv`, and `annotations.csv` (schema `s4.dataset-manifest.v2`). Source images and case records are not changed. Untouched AI, human-authored, corrected, removed, and CVAT-imported states remain distinguishable in the export.

10 Models & Audit and Traceability

10.1 Models & Audit

Open **Models & Audit** when technical inspection is needed. It is read-only and is not another review queue. For the selected case it shows:

- model readiness, model IDs, revisions, runtime state, and limitations;
- the RETFound score context and the PRISM-DR lesion counts by class;
- **Case lineage & annotation provenance**: source SHA-256, analysis SHA-256, AI visual evidence, human annotations, CVAT imports, and inference metadata.

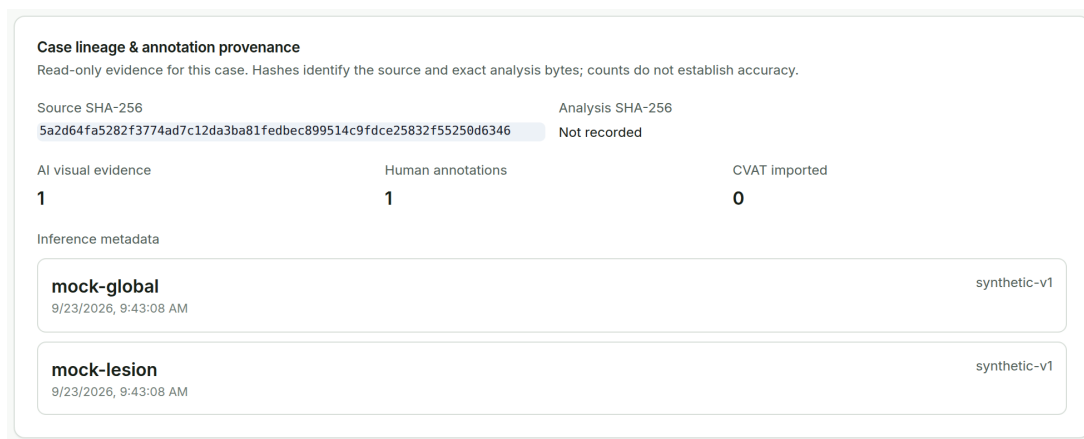


Figure 10.1: Case lineage for SYNTH_001. The synthetic fixture uses the explicit mock providers, so no analysis derivative is recorded.

10.2 What the workstation records

For every case the workstation keeps:

Record	Contents
Source identity	Original filename, source SHA-256, admission state; the source file is never modified.
Case context	Pseudonymous patient key, eye, reviewer, and timestamp from Confirm Image .
Model evidence	Model ID, revision, preprocessing, analysis SHA-256, grade suggestion, raw lesion output.
Clinician decisions	Final DR grade, review action (AI_ACCEPTED, AI_CORRECTED, MANUAL), remark, reviewer, time.
Annotations	Human annotations, AI ROI corrections and removals with the original AI evidence, and the confirmation hash.
Dataset state	DR-ready and Lesion-ready status and the exported manifest.

10.3 Optional CVAT

CVAT Online can be used for dense annotation or geometry correction when your operator has configured it. The round trip preserves source and prediction hashes. CVAT is optional and is not needed for offline review or export; the local workstation remains the source of the final case state.

11 Troubleshooting, Safety, and Glossary

11.1 Troubleshooting

What you see	What to do
Unsupported modality	The DICOM is recognized but is not supported fundus input. Do not send it to the models.
AI analysis is not available	Continue image review and human decisions. Ask the operator to check the Model API and the workstation connection.
No PRISM overlays	Check Show AI suggestions , the lesion filter, and whether a result exists. An empty result does not prove lesion absence.
Save failed — retry	Stay on the page, check the reviewer name and connection, then choose Save draft again.
Case is not DR-ready	Check Confirm Image , the final grade, the reviewer, and the timestamp.
Case is not Lesion-ready	Check whether the annotation set changed after Confirm Annotation ; confirm it again.
Wrong reviewer prefilled	Edit the field, or change Default reviewer in Settings .
CVAT is unavailable	Continue local review. Retry once the operator confirms URL, project, credentials, and network.

11.2 Safety and privacy

Safety – clinical use

This is a research and clinical-support proof of concept. Model output is assistive evidence and human review remains authoritative. The product does not establish diagnosis, referral, treatment, or disease truth.

Safety – data handling

Use only approved public, synthetic, or authorized data. Never put PHI, private DICOM headers, tokens, passwords, CVAT credentials, model weights, or runtime databases in screenshots, examples, exports, or source control.

11.3 Glossary

Term	Meaning
AI suggestion	A model output shown to support inspection.
Confirmation milestone	One of Confirm Image , Confirm DR Grade , Confirm Annotation .
DR-ready	The case meets the dataset rules for the DR-grade task.
Lesion-ready	The case meets the dataset rules for the lesion-annotation task.
Model score	A value returned by a model; not a calibrated clinical probability.
Original AI evidence	Class, score, geometry, identity, model, and provenance returned by inference, kept unchanged after human edits.
Reviewer default	A browser-local name used to prefill new forms; not authentication.